Algorithms for Uncertainty Quantification

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1. Introduction to UQ

Definition 1.1. UQ = (probability theory + statistical practice) + “real world”

- UQ = Large-scale computational problems.
- UQ = Challenging; smart techniques are needed to overcome the computational challenges.
- In lecture, we focus on (but don’t limit ourselves to) forward propagation.

2. Repetition of probability theory and statistics

2.1. Two ways to think about probability

Before introducing the formal definitions, it helps to look at two major interpretations of probability.

The **frequentist** view says that probability is the long-run relative frequency of an event if you repeat the same experiment many times.

The **Bayesian** view says that probability is expresses a degree of belief, which can be updated when new data becomes available.

These are not just abstract philosohpies. They influence how we interpret uncertain quantities.

In a frequentist mindset, saying "the probability of heads is $\frac{1}{2}$" means that if you flip a fair coin many times, about half of the outcomes will be heads.

In a Bayesian mindset, probability can describe uncertainty in unknown quantities themselves, such as an uncertain model parameter.

For uncertainty quantification, both viewpoints matter. Frequentist tools help estimate uncertainty from data, while Bayesian tools help update uncertainty when new evidence is observed.

2.2. Start with the simplest experiment: Coin Flipping

Let us begin with the most elementary example: flipping coings.

For one fair coin, the possible outcomes are:
- head
- tail

Since the coin is fair, each outcome has probability $\frac{1}{2}$.

Now consider **two fair coins flipped simultaneously**. The possible outcomes are

$$(h,h),\ (t,h),\ (h,t),\ (t,t)$$

Each outcome is equally likely, so each one has probability $\frac{1}{4}$.

Now suppose I ask for the probability of these two events:

$$A = \{(h,h),(t,h)\}$$

and

$$B = \{(h,h),(h,t)\}$$

Each event contains 2 of the 4 equally likely outcomes, so

$$P(A) = \frac{1}{2}, \quad P(B) = \frac{1}{2}$$

This example teaches an important lesson:

probability questions are usually about sets of outcomes, not just individual outcomes.

That is exactly why we need the formal concept of a probability space.

2.3. Probability space: The formal foundation

A probability space is a triple

$$(\Omega, \mathcal{F}, P)$$

where:

- Ω is the **sample space**, the set of all possible outcomes.
- $\mathcal{F}$ is a **σ-algebra**, the collection of events.
- P is a **probability measure**, which assigns a number between 0 and 1 to each event.

The probability measure P satisfies three axioms:

1. $P(\emptyset) = 0$
2. $P(\Omega) = 1$
3. For pairwise disjoint events A_i,

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i)$$

The third property is called **countable additivity**. It says that if events cannot happen simultaneously, then the probability that one of them happens is the sum of their probabilities.

2.3.1. Why do we need $\mathcal{F}$

In simple finite examples like coin tossing, $\mathcal{F}$ is usually just the set of all subsets of Ω. But in continuous settings, not every imaginable subset can be handled cleanly. The σ-algebra tells us which sets are measurable and therefore allowed to be assigned probabilities.

2.3.2. Coin example as a probability space

For one fair coin:

- $\Omega = \{h, t\}$
- $\mathcal{F} = \{\emptyset, \{h\}, \{t\}, \{h, t\}\}$
- $P(\{h\}) = P(\{t\}) = \frac{1}{2}$

For two fair coins:
- $\Omega = \{(h,h),(h,t),(t,h),(t,t)\}$
- $\mathcal{F}$ consist of the events built from these outcomes
- $A = \{(h,h),(t,h)\}$ has probability $\frac{1}{2}$
- $B = \{(h,h),(h,t)\}$ also has probability $\frac{1}{2}$

So the first major idea is:

probability is defined on events inside a probability space.

2.4. Random variables: turning outcomes into numbers

Once the probability space is in place, the next step is to introduce the central object of statistics: the **random variable**.

A random variable is a function

$$X : \Omega \to \mathbb{R}$$

with the property that sets of the form

$$\{\omega \in \Omega \mid X(\omega) \leq x\}$$

are events.

The formal definition is technical, but the intuition is simple:

A random variable takes a possibly complicated outcome and assigns it a number.

For the two-coin experiment, define $X(\omega)$ as the **number of tails**. Then
- $X(h,h) = 0$
- $X(h,t) = 1$
- $X(t,h) = 1$
- $X(t,t) = 2$

So instead of working with the raw outcomes $(h,h),(h,t)$, and so on, we work with the numerical quantity “number of tails”.

This is exactly what makes probability useful in applications. Real examples of random variables are:

- tomorrow’s temperature
- the number of defective parts in a batch
- the waiting time in a queue
- the pressure at a measurement point
- the output of a model with uncertain inputs

A random variable is called **discrete** if it takes values in a countable set, and **continuous** otherwise.

The number of tails is discrete because it can only take the values 0, 1, 2. A quantity like time or height is often modeled as continuous.

2.5. The CDF: one function that works for every random variable

For any random variable X, I define a **cumulative distribution function** by

$$F_X(x) = P(X \leq x)$$

This is the distribution function attached to every random variable.

That universality is important: the CDF always exists, where X is discrete, continuous, or mixed.

For the random variable X that counts the number of tails, the CDF is

$$F_X(x) = \begin{cases} 0, & x < 0 \\ \frac{1}{4}, & 0 \leq x < 1 \\ \frac{3}{4}, & 1 \leq x < 2 \\ 1, & x \geq 2 \end{cases}$$

Here is the logic:
- if $x < 0$, then $X \leq x$ is impossible
- if $0 \leq x < 1$, then only the case $X = 0$ is included, so the probability is $\frac{1}{4}$
- if $1 \leq x < 2$, then the cases $X = 0$ and $X = 1$ are included, so the probability is $\frac{1}{4} + \frac{1}{2} = \frac{3}{4}$
- if $x \geq 2$, then all possible values are included, so the probability is 1

This step-like behaviour is typical for discrete random variables.

2.5.1. Intuition for the CDF

A CDF answers the question:

How much probability has accumulated up to this point?

As x moves from left to right, the CDF keeps collecting probability.

2.6. Continuous random variables and the PDF

Now let us specialize to continuous random variables.

A random variable is called continuous ift its CDF can be written as

$$F_X(x) = \int_{-\infty}^{x} f_X(s)ds$$

The function f_X is the **probability density function**, or PDF, and it satisfies

$$f_X(x) = \frac{dF_X(x)}{dx}$$

with

$$f_X : \mathbb{R} \to [0, \infty)$$

This leads to a very important distinction:
- the **CDF** gives probabilities directly through $P(X \leq x)$
- the **PDF** does not directly give probabilities at individual points
- probbilities come from **areas under the density**

Specifically,

$$P(a \leq X \leq b) = \int_a^b f_X(x)dx$$

2.6.1. Support of a PDF

The **support** of the density is the set where it is nonzero:

$$\text{supp}(f_X) = \{x \in \mathbb{R} \mid f_X(x) \neq 0\}$$

Two key properties are:
- $f_X(x) \geq 0$
- the total integral equals 1:

$$\int_{\text{supp}(f_X)} f_X(x)dx = 1$$

That second condition expresses the fact that the total probability must be 1.

2.6.2. An important interpretation

For a continuous random variable, the probability of taking one exact value is usually zero:

$$P(X = x) = 0$$

That can feel strange at first. The density may be high at a point, but the probability of a single exact point is still zero because probability comes from area, not from height alone.

2.7. Expectation and variance: the first two key summaries

Once a distribution is defined, the next question is: how do I summerize it? The two most important summaries are the **expectation** and the **variance**.

2.7.1. Expectation

The expectation, or mean is

$$\mu = E[X] = \int_{\mathrm{supp}(f_X)} x f_X(x) dx$$

It describes the center or average value of the distribution.

2.7.2. Variance

The variance is

$$\sigma^2 = \mathrm{Var}(X) = \int_{\mathrm{supp}(f_X)} (x - E[X])^2 f_X(x) dx$$

An equivalent and very useful formula is

$$\mathrm{Var}(X) = E[X^2] - E[X]^2$$

Variance measures how strongly the values of X are spread around the mean.

2.7.2.1. Intuition

- large variance means the values are widely spread
- small variance means the values are tightly clustered near the mean

In many applications, these two number already tell use a lot about the uncertainty: where it is centered and how strong the spread is.

2.8. Discrete random variables and the PMF

For a discrete random variable, the natural object is the **probability mass function**, or PMF:

$$f_X(x) = P(X = x)$$

This is the discrete analogue of the PDF, but the interpretation is different.

- A **PMF** assigns probability directly to individual points.

- A PDF assigns density, and probabilities come from integrals.

For the two-coin example with $X =$ number of tails,
- $P(X = 0) = \frac{1}{4}$
- $P(X = 1) = \frac{1}{2}$
- $P(X = 2) = \frac{1}{4}$

That is the PMF of X

Now suppose I observe realizations $X_1, \ldots, X_n$. Then I can compute the **sample mean**

$$\hat{X} = \frac{1}{n} \sum_{i=1}^{n} X_i$$

and the **sample variance**

$$S^2 = \frac{1}{n-1} \sum_{i=1}^{n} \left(X_i - \hat{X} \right)^2$$

Notice the denominator $n-1$, not n. That detail matters in statistics.

2.8.1. Why this matters

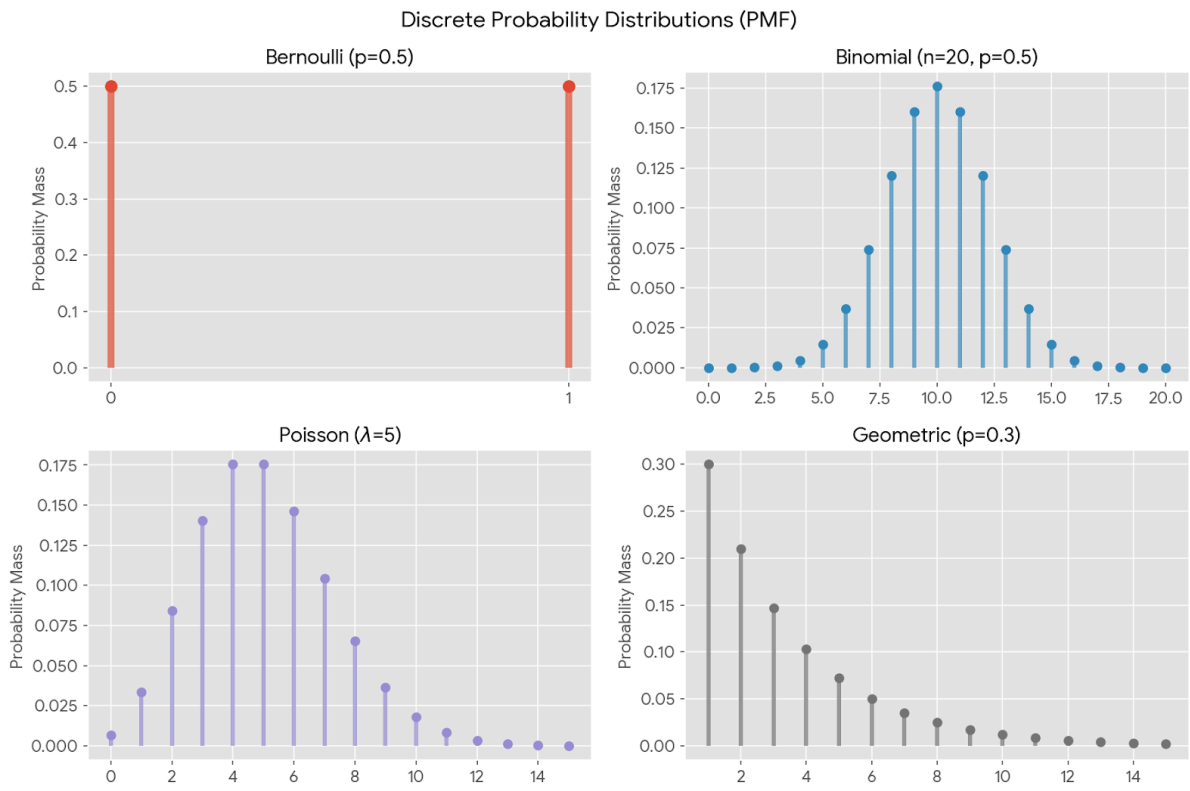
This is where theort meets data:
- $E[X]$ and $\mathrm{Var}(X)$ are population quantities
- $\hat{X}$ and S^2 are estimates based on a sample.

That distinction is central in uncertainity quantification because the true distribution is often unknown and has to be inferred from data.

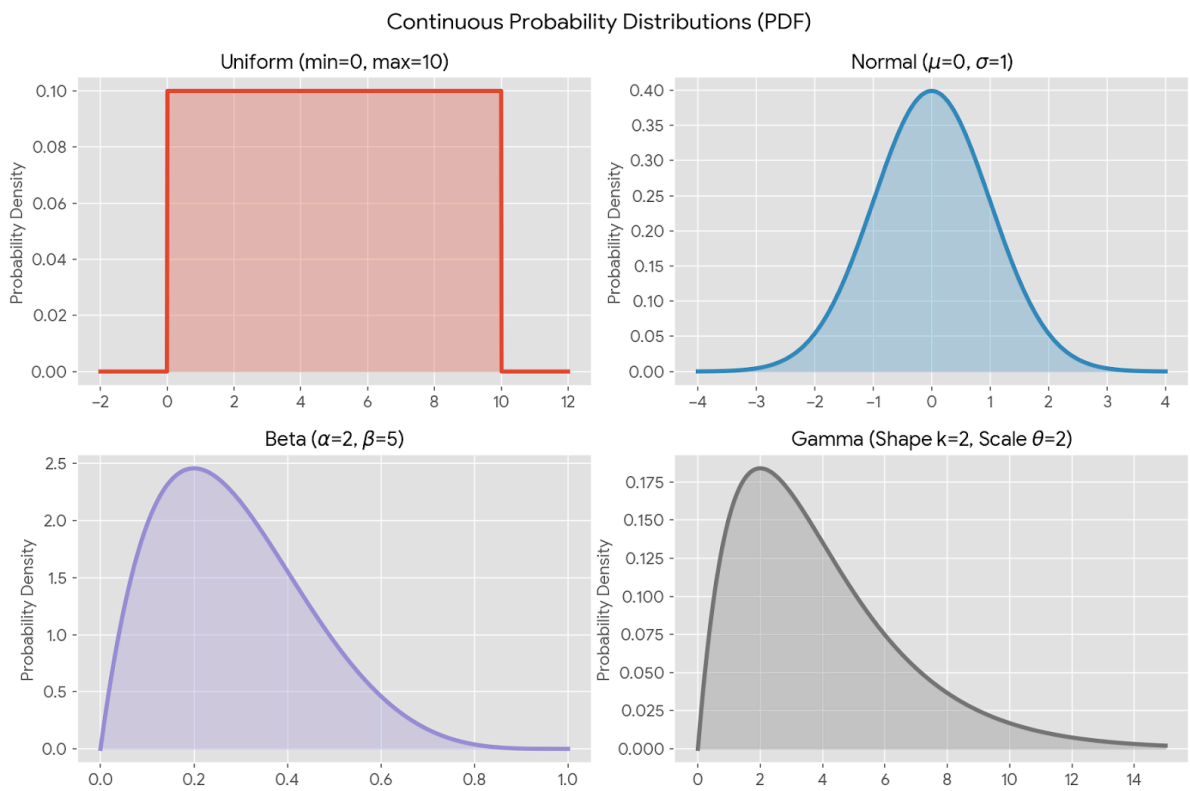
2.9. Examples of common distributions

Some standard distributions appear over and over again.

For discrete random variables, common examples are:



For continuous random varialbes, common examples are:



Among these, the normal and uniform distributions are especially important because they are simple, useful, and easy to interpret.

2.10. The Normal Distribution

A normally distributed random variable has density

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

and I write

$$X \sim \mathcal{N}(\mu, \sigma^2)$$

to mean that X is normal with mean μ and variance σ^2.

The special case

$$\mathcal{N}(0, 1)$$

is called the **standard normal ditribution**

2.10.1. What do μ and σ do?

- μ shifts the bell curve left or right
- σ controls the spread

More precisely:
- larger σ: wider and flatter curve
- smaller σ: narrower and taller curve

The height changes because the total area under the density must always remain 1.

2.10.2. Why is the normal distribution so important?

The normal distribution appears everywhere because sums and averages of many small independent effects often behave approximately normally. That is why Gaussian models are used for measurement noise, physical fluctuations, modeling errors, and uncertain parameters.

2.11. The Uniform Distribution

The uniform distribution on the interval $[a, b]$ has density

$$f_X(x) = \frac{1}{b-a} I_{[a,b]}(x)$$

where the indicator function $I_{[a,b]}(x)$ is 1 if $x \in [a, b]$ and 0 otherwise.

I write

$$X \sim U(a, b)$$

for a uniform random variable on $[a, b]$.

Its mean and variance are

$$E[X] = \frac{a+b}{2}, \quad \mathrm{Var}(X) = \frac{(b-a)^2}{12}$$

2.11.1. Interpretation

The uniform distribution says that every location in the interval is equally likely in the density sense. There is no preffered point inside $[a, b]$.

If the interval becomes wider, the density gets lower, because the total area under the density still has to remain 1.

2.11.2. When do we use uniform distributions?

A uniform distribution often models **bounded uncertainty with no preference**. Typical examples are:
- a parameter known only to lie between two limits
- an initialization range
- a rough prior model when little else is known

2.12. Moving from one variable to many: Random Vectors

So far I have worked with one random variable at a time. Now I move to several uncertain quantities simultaneously.

A random vector is a mapping

$$X : \Omega \to \mathbb{R}^n, \quad X = (X_1, ..., X_n)^T$$

where each component X_i is a random variable.

This is how I model several uncertain quantities together, such as:
- temperature, pressure, and velocity
- several uncertain parameters in one model
- a position in 2D or 3D
- the full state of a system

As soon as there is more than one variable, I care not only about each one separately, but also about how they depend on one another.

2.13. Covariance and correlation

For two random variables X and Y, the covariance is

$$\mathrm{cov}(X, Y) = E[(X - E[X])](Y - E[Y])$$

and equivalently

$$\mathrm{cov}(X, Y) = E[XY] - E[X]E[Y]$$

The Pearson correlation coefficient is

$$\rho_{XY} = \frac{\mathrm{cov}(X,Y)}{\sqrt{\mathrm{Var}(X)\mathrm{Var}(Y)}} \quad \text{with} \quad \rho_{XY} \in [-1,1]$$

2.13.1. What covariance tells us

Covariance measures whether two variables tend to move together.

- positive covariance: when one is above its mean, the other also tends to be above its mean.
- negative covariance: when one is above its mean, the other tends to be below its mean.
- zero covariance: no linear co-movement is detected.

2.13.2. Why correlation is often preferred?

Covariance depends on units. If I change the units of measurement, the covariance changes scale as well.

Correlation removes that scale dependence and always lies between -1 and 1:

- $\rho = 1$: perfect positive linear relation
- $\rho = -1$: perfect negative linear relation
- $\rho = 0$: no linear correlation

But I have to be careful here:

zero correlation does not necessarily mean independence.

Correlation only captures linear dependence. Independence is a stronger concept.

2.14. Expectation and variance of sums

One of the most useful facts in probability is that expectation is linear.

For constants $a_1, \ldots, a_n$

$$E\left[\sum_{i=1}^{n} a_i X_i\right] = \sum_{i=1}^{n} a_i E[X_i]$$

This property is extremely important in uncertainty quantification.

For variance, the corresponding formula is

$$\mathrm{Var}\left(\sum_{i=1}^{n} a_i X_i\right) = \sum_{i=1}^{n} a_i^2 \ \mathrm{Var}(X_i) + 2\sum_{i<j} a_i a_j \ \mathrm{cov}(X_i, X_j)$$

If the variables are uncorrelated, all covariance terms vanish, and this simplifies to

$$\operatorname{Var}\left(\sum_{i=1}^{n} a_i X_i\right) = \sum_{i=1}^{n} a_i^2 \operatorname{Var}(X_i)$$

2.14.1. Why this matters in uncertainty quantification

In UQ, I constantly ask how uncertainty in the inputs propagates to the output.

These formulas show:

- how the mean of an output depends on the means of the inputs.
- how the variance of an output depends not only on the input variances, but also on the dependence between inputs

So the key lesson is:

dependence matters.

Even if two variables have the same individual variances, the variance of their sum can change substantially depending on whether they are positively correlated, negatively correlated, or uncorrelated.

2.15. Independence

For two events A and B, the join probability is

$$P(A, B) = P(A \cap B) = P(A \mid B)P(B)$$

I say that A and B are independent if

$$P(A, B) = P(A)P(B)$$

Equivalently,

$$P(A \mid B) = P(A)$$

So independence means:

knowing that B occured tells me nothing about whether A occured.

For random variables, independence is expressed by factorization of the joint density or mass function:

$$f_{XY}(x, y) = f_X(x) f_Y(y)$$

2.15.1. Independence vs. uncorrelatedness

This distinction is crucial:

- independence often implies zero covariance
- zero covariance does **not** generally imply independence

uncorrelatedness only says there is no linear dependence. Independence says the entire joint distribution factorizes.

2.16. i.i.d random variables

A collection $X_1, ..., X_n$ is called **independent and identification distributed**, or **i.i.d.** if:
1. the variables are mutually independent
2. they all have the same distribution
3. the joint density factorizes as

$$f(x_1, ..., x_n) = \prod_{i=1}^{n} f_X(x_i)$$

This assumption is one of the most important in all of probability and statistics.

2.16.1. Why i.i.d matters so much

The i.i.d. framework underlies:
- sample averages
- Monte Carlo simulation
- many estimators
- the law of large numbers
- the central limit theorem

In uncertainty quantification, when I generate many samples of uncertain inputs, the i.i.d. assumotion is often what makes convergence analysis possible.

2.17. The mutlivariate normal distribution

The multivariate extension of the Gaussian is one of the most important models for joint uncertainty.

An n-dimensional random vector is normally distributed with mean vector

$$\mu = (\mu_1, ..., \mu_n)^T$$

and covariance matrix V, where

$$V_{ij} = \mathrm{cov}\big(X_i, X_j\big)$$

I write

$$X \sim \mathcal{N}(\mu, V)$$

and its density is

$$f_X(x) = \frac{1}{\sqrt{(2\pi)^n |V|}} \exp\Big(-\frac{1}{2}(x-\mu)^T V^{-1}(x-\mu)\Big)$$

The **standard multivariate normal** has zero mean and identity covariance matrix.

2.17.1. What the different pieces mean

- μ gives the center of the probability cloud
- V determines the spread in each direction and the correlation structure
- $|V|$ is the determinant, which appears in the normalization
- V^{-1} shapes the quadratic form in the exponential, which controls how fast the density decays away from the mean

In two dimensions standard bivariate normal looks like a smooth bell-shaped surface centered at (0,0). The density is highest at the center and decreases symmetrically as I move away from the origin.

This is simply the multidimensional version of the familiar one-dimensional bell curve.

2.17.2. Random Variables

2.17.2.1. CDF - Cumulative Distribution Function

2.17.2.2. PDF - Probability Density Function

2.17.3. Expectation, Variance derivation is an exam question

How to derive the variance basically.

Slides 20 deriving expectation is exampl question. Exam question again deriving variance from expectations -> $\mathbb{E}[X^2] - \mathbb{E}[X]^2 = \mathrm{Var}(X)$

Exam question deriving covariance as well as pearson coeffecient.